Enhancing Reasoning Capabilities of LLMs via Principled Synthetic Logic Corpus

This presentation summarizes the NeurIPS 2024 paper on improving large language models' reasoning.

We introduce Additional Logic Training (ALT) using a synthetic logic corpus.

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Introduction and Motivation

LLM Reasoning Limits

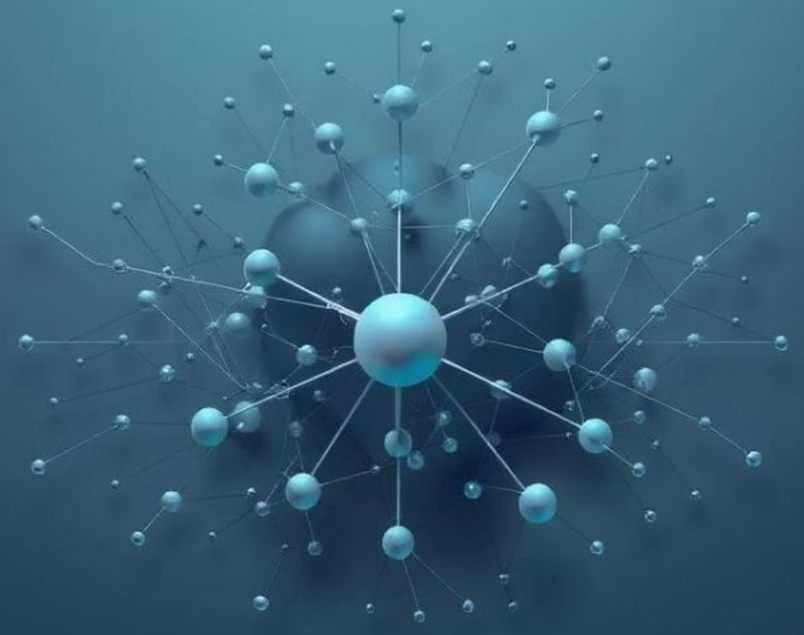
LLMs memorize over reasoning and falter on new logic problems.

Challenges

Training corpora contain logical fallacies and biases, impairing robust reasoning.

Solution: Additional Logic Training (ALT)

ALT employs a synthetic corpus focused on core deductive reasoning



Related Work and Contributions

Prior Work Limitations

- RuleTaker: Narrow rules, limited vocabulary
- PARARULE-Plus: Limited diversity
- FLD: Smaller vocab and complexity
- NeuBAROCO: Focused on syllogisms

Our Contributions

- Formal design rooted in symbolic logic theory
- Large, diverse corpus with 100k unique samples
- Generalizes to math, code, and NLI tasks

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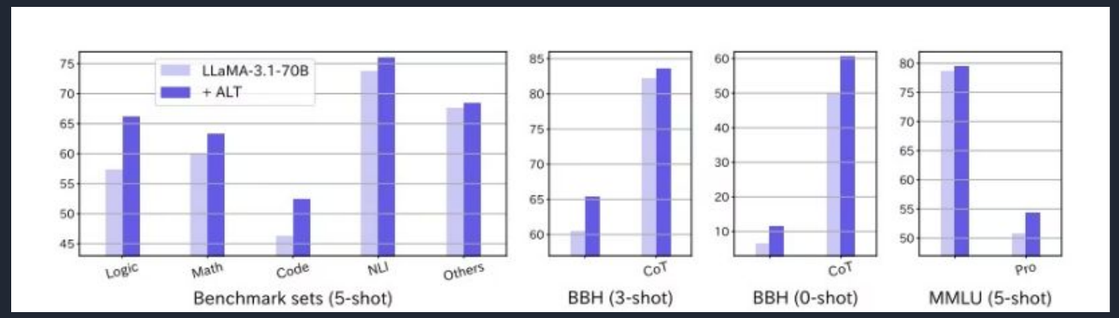

LLama 3.1 vs +ALT

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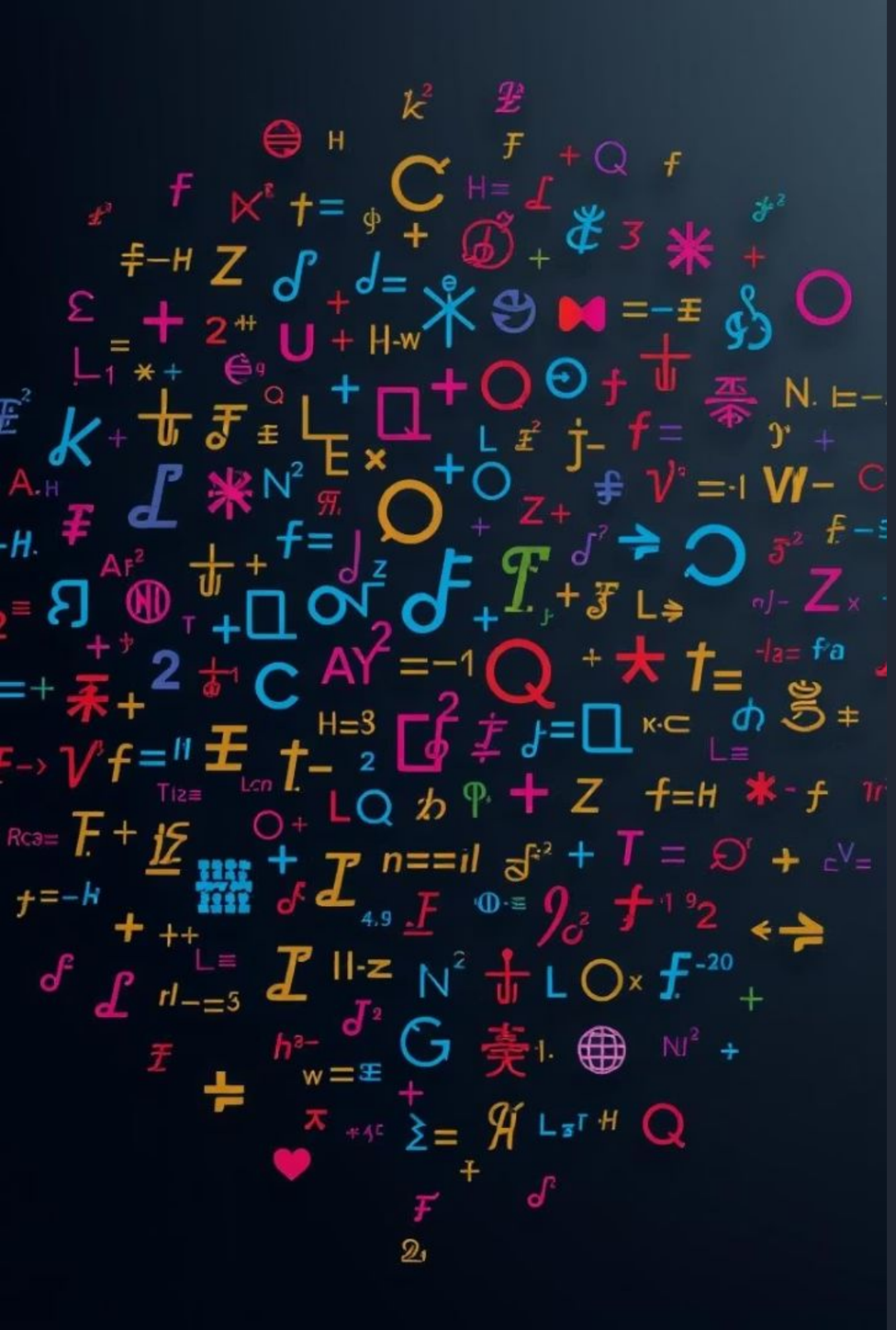

Design Principles for Synthetic Logic Corpus

DP1: Unknown Facts

Ensure reasoning beyond factual knowledge using random symbols.

DP2: Illogical Reasoning

Include adversarial distractors to identify invalid inferences.

DP3: Diverse Reasoning Rules

Cover ~50 logic rules including axioms and De Morgan's laws.

DP4: Linguistic Expressions

Use varied templates to prevent overfitting to phrasing.

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Comparison on different corpora

Table 1: Synthetic logic corpora compared in this study, with their features categorized according to our proposed design principles (DP). Note that the last row of the *ablation* corpora lists variations of $\text{FLD}_{\times 2}$, each of which differs from the original regarding one of the design principles.

	DP1	DP2	DP3		DP4
	vocabulary size	distractors	deduction rules	logical steps	expressions per formula
RuleTaker (Clark et al., 2021) (RT)	≤ 100 (hand-selected)	random formula	2 (implication)	1–5	$\mathcal{O}(1)$
PARARULE-Plus (Bao et al., 2022) (PRP)	≤ 100 (hand-selected)	random formula	2 (implication)	1–5	$\mathcal{O}(1)$
FLD (Morishita et al., 2023)	$\simeq 15$k (WordNet, subset)	random formula	13 (axioms)	**1–8**	10∼100
$\text{FLD}_{\times 2}$	**$\simeq 100$k (WordNet, full)**	**adversarial formula**	**$\simeq 50$ (axioms and theorems)**	**1–8**	**10∼100 (more extensive than FLD)**
$\text{FLD}_{\times 2}$ *ablation* corpora →	100 → *w/o DP1*	not used → *w/o DP2*	2 (implication) → *w/o DP3.rules*	1 → *w/o DP3.steps*	1 → *w/o DP4*

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FLD×2 Corpus Construction

Vocabulary and Tokens

Approx. 100k tokens from WordNet vocabulary.

Logical Rules and Samples

~50 rules, 100,000 multi-step deduction samples with varied templates.

Adversarial Elements

0–20 distractors per sample to challenge reasoning.

Sample Composition

Each sample has facts, hypothesis, and 1-8 logical steps.

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Experimental Setup

Models and Training LLaMA-3.1 8B & 70B versions Fine-tuned 1 epoch on...

Experimental Setup

Models and Training

- LLaMA-3.1 8B & 70B versions
- Fine-tuned 1 epoch on 100k samples
- Batch size 256; learning rates 2e-5 & 3e-6
- Recall Adam optimizer

Benchmarks

- Logic: FOLIO, LogicNLI, AbductionRules
- Math: GSM8k, MathQA
- Code: HumanEval, MBPP, MultiPL-E
- NLI: SNLI, RTE, MNLI
- Others: CommonsenseQA, HellaSwag, SciQ, ARC

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Experimental Results

LLaMA-3.1-8B Improvements

Accuracy rises from 47.9% to 52.0% overall.

LLaMA-3.1-70B Gains

Overall from 60.0% to 64.4%; logic from 57.4% to 66.1%.

Cross-Domain Boosts

Notable gains in Math, Code, NLI, and other tasks.

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Ablation Study Results

The study highlights the impact of each design principle on model performance.

Removing DP1 or DP2 severely reduces overall accuracy.

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Enhancing LLMs with Additional Logic Training

The table shows the performance of LLMs before and after ALT. **ALT significantly improves performance** of LLMs across the board.

Even **LLaMA-3.1-70B**, trained on 15T+ tokens, gained **up to 10 points**.

Confirms ALT's effectiveness on **state-of-the-art LLMs**.

ALT-FLD×2 model (trained on the full FLD×2 corpus with all design principles) achieved the **best generalization performance**.

(a) LLaMA-3.1-8B.

	Avg.	Logic	Math	Code	NLI	Others	BBH (3-shot)		BBH (0-shot)		MMLU	
								CoT		CoT		Pro
LLaMA-3.1-8B	47.9	42.8$_{\pm 0.4}$	39.6$_{\pm 0.5}$	35.4	65.4$_{\pm 0.3}$	60.7$_{\pm 0.3}$	44.9$_{\pm 0.4}$	61.9$_{\pm 0.4}$	8.2$_{\pm 0.2}$	36.5$_{\pm 0.4}$	65.3$_{\pm 0.4}$	35.8$_{\pm 0.4}$
⊕ALT-PRP	48.1	43.7$_{\pm 0.2}$	39.2$_{\pm 0.3}$	35.7	65.6$_{\pm 0.2}$	60.8$_{\pm 0.2}$	44.9$_{\pm 0.2}$	61.8$_{\pm 0.2}$	8.2$_{\pm 0.1}$	36.4$_{\pm 0.2}$	65.3$_{\pm 0.2}$	35.3$_{\pm 0.2}$
⊕ALT-RT	50.1	46.8$_{\pm 0.1}$	42.4$_{\pm 0.2}$	36.5	68.6$_{\pm 0.1}$	61.3$_{\pm 0.1}$	46.9$_{\pm 0.2}$	63.5$_{\pm 0.2}$	13.7$_{\pm 0.1}$	38.4$_{\pm 0.2}$	65.3$_{\pm 0.1}$	35.7$_{\pm 0.2}$
⊕ALT-FLD	51.9	51.6$_{\pm 0.1}$	43.4$_{\pm 0.2}$	38.1	70.1$_{\pm 0.1}$	61.5$_{\pm 0.1}$	46.7$_{\pm 0.2}$	64.9$_{\pm 0.2}$	11.9$_{\pm 0.1}$	39.6$_{\pm 0.2}$	65.4$_{\pm 0.1}$	36.2$_{\pm 0.2}$
⊕**ALT-FLD**$_{\times 2}$	52.0	52.2$_{\pm 0.1}$	43.2$_{\pm 0.2}$	38.0	70.7$_{\pm 0.1}$	61.5$_{\pm 0.1}$	46.5$_{\pm 0.2}$	65.3$_{\pm 0.2}$	11.3$_{\pm 0.1}$	38.7$_{\pm 0.2}$	65.5$_{\pm 0.1}$	36.4$_{\pm 0.2}$

(b) LLaMA-3.1-70B.

	Avg.	Logic	Math	Code	NLI	Others	BBH (3-shot)		BBH (0-shot)		MMLU	
								CoT		CoT		Pro
LLaMA-3.1-70B	60.0	57.4$_{\pm 0.4}$	60.0$_{\pm 0.5}$	46.2	73.7$_{\pm 0.3}$	67.7$_{\pm 0.3}$	60.4$_{\pm 0.3}$	82.1$_{\pm 0.2}$	6.5$_{\pm 0.1}$	50.1$_{\pm 0.3}$	78.7$_{\pm 0.3}$	50.7$_{\pm 0.4}$
⊕ALT-PRP	60.4	57.7$_{\pm 0.4}$	59.8$_{\pm 0.5}$	49.2	73.5$_{\pm 0.3}$	67.6$_{\pm 0.3}$	60.4$_{\pm 0.4}$	82.2$_{\pm 0.3}$	6.0$_{\pm 0.2}$	50.1$_{\pm 0.4}$	78.7$_{\pm 0.3}$	50.9$_{\pm 0.4}$
⊕ALT-RT	62.7	61.4$_{\pm 0.2}$	62.1$_{\pm 0.3}$	50.8	75.4$_{\pm 0.2}$	68.4$_{\pm 0.2}$	64.1$_{\pm 0.3}$	82.5$_{\pm 0.2}$	11.5$_{\pm 0.2}$	59.2$_{\pm 0.3}$	79.0$_{\pm 0.2}$	52.4$_{\pm 0.3}$
⊕ALT-FLD	64.2	65.7$_{\pm 0.1}$	63.6$_{\pm 0.2}$	52.0	75.3$_{\pm 0.1}$	68.5$_{\pm 0.1}$	65.0$_{\pm 0.2}$	83.6$_{\pm 0.1}$	12.1$_{\pm 0.1}$	59.9$_{\pm 0.2}$	79.3$_{\pm 0.1}$	54.4$_{\pm 0.2}$
⊕**ALT-FLD**$_{\times 2}$	64.4	66.1$_{\pm 0.1}$	63.3$_{\pm 0.2}$	52.4	76.1$_{\pm 0.1}$	68.5$_{\pm 0.1}$	65.4$_{\pm 0.2}$	83.6$_{\pm 0.2}$	11.4$_{\pm 0.1}$	60.8$_{\pm 0.2}$	79.5$_{\pm 0.1}$	54.4$_{\pm 0.2}$

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Enhancing LLMs with Additional Logic Training (continued)

Ablation results:

Removing any single design principle from FLD×2 leads to **performance drops**.

Every design principle is **essential** to maximize ALT's benefit.

Table 3: LLaMA-3.1-8B trained on the ablation corpora.

	Avg.	Logic	Math	Code	NLI	Others	BBH (3-shot)		BBH (0-shot)		MMLU	
								CoT		CoT		Pro
⊕**ALT-FLD**$_{\times 2}$	**52.0**	**52.2**$_{\pm 0.1}$	**43.2**$_{\pm 0.2}$	38.0	70.7$_{\pm 0.1}$	**61.5**$_{\pm 0.1}$	46.5$_{\pm 0.2}$	**65.3**$_{\pm 0.2}$	11.3$_{\pm 0.1}$	38.7$_{\pm 0.2}$	65.5$_{\pm 0.1}$	**36.4**$_{\pm 0.2}$
w/o DP1	51.4	**52.2**$_{\pm 0.1}$	43.1$_{\pm 0.2}$	**39.2**	70.0$_{\pm 0.1}$	59.4$_{\pm 0.1}$	46.7$_{\pm 0.2}$	64.7$_{\pm 0.2}$	11.5$_{\pm 0.1}$	38.9$_{\pm 0.2}$	65.4$_{\pm 0.1}$	36.1$_{\pm 0.2}$
w/o DP2	50.6	49.9$_{\pm 0.1}$	43.1$_{\pm 0.2}$	38.1	**71.1**$_{\pm 0.1}$	59.3$_{\pm 0.1}$	46.1$_{\pm 0.2}$	64.6$_{\pm 0.2}$	10.4$_{\pm 0.1}$	37.4$_{\pm 0.2}$	65.4$_{\pm 0.1}$	35.7$_{\pm 0.2}$
w/o DP3.rules	50.7	50.4$_{\pm 0.1}$	42.8$_{\pm 0.2}$	38.3	69.5$_{\pm 0.1}$	59.4$_{\pm 0.1}$	46.4$_{\pm 0.2}$	64.0$_{\pm 0.2}$	11.8$_{\pm 0.1}$	38.3$_{\pm 0.2}$	**65.6**$_{\pm 0.1}$	36.2$_{\pm 0.2}$
w/o DP3.steps	51.1	51.5$_{\pm 0.1}$	43.1$_{\pm 0.2}$	38.7	69.6$_{\pm 0.1}$	59.5$_{\pm 0.1}$	**46.8**$_{\pm 0.2}$	65.0$_{\pm 0.2}$	12.3$_{\pm 0.1}$	38.8$_{\pm 0.2}$	**65.6**$_{\pm 0.1}$	36.3$_{\pm 0.2}$
w/o DP4	51.3	**52.2**$_{\pm 0.1}$	42.8$_{\pm 0.2}$	38.4	70.3$_{\pm 0.1}$	59.5$_{\pm 0.1}$	46.1$_{\pm 0.2}$	64.8$_{\pm 0.2}$	**12.8**$_{\pm 0.1}$	**39.3**$_{\pm 0.2}$	65.5$_{\pm 0.1}$	36.3$_{\pm 0.2}$

Knowledge forgetting:

Models trained **without Recall Adam** (no forgetting prevention) performed worse — even worse than LLMs **without ALT**.
Preventing knowledge forgetting is **critical** to successful ALT training.

(a) LLaMA-3.1-8B.

	Avg.	Logic	Math	Code	NLI	Others	BBH (3-shot)		BBH (0-shot)		MMLU	
								CoT		CoT		Pro
LLaMA-3.1-8B	47.9	42.8$_{\pm 0.4}$	39.6$_{\pm 0.5}$	35.4	65.4$_{\pm 0.3}$	60.7$_{\pm 0.3}$	44.9$_{\pm 0.4}$	61.9$_{\pm 0.4}$	8.2$_{\pm 0.2}$	36.5$_{\pm 0.4}$	65.3$_{\pm 0.4}$	35.8$_{\pm 0.4}$
⊕ALT-PRP$_{\text{w/o RecAdam}}$	43.5	39.5$_{\pm 0.2}$	29.1$_{\pm 0.3}$	35.3	57.8$_{\pm 0.2}$	61.0$_{\pm 0.2}$	40.5$_{\pm 0.2}$	47.0$_{\pm 0.2}$	3.9$_{\pm 0.1}$	6.3$_{\pm 0.1}$	64.9$_{\pm 0.2}$	34.0$_{\pm 0.2}$
⊕ALT-PRP	48.1	43.7$_{\pm 0.2}$	39.2$_{\pm 0.3}$	35.7	65.6$_{\pm 0.2}$	60.8$_{\pm 0.2}$	44.9$_{\pm 0.2}$	61.8$_{\pm 0.2}$	8.2$_{\pm 0.1}$	36.4$_{\pm 0.2}$	65.3$_{\pm 0.2}$	35.3$_{\pm 0.2}$
⊕ALT-RT	50.1	46.8$_{\pm 0.1}$	42.4$_{\pm 0.2}$	36.5	68.6$_{\pm 0.1}$	61.3$_{\pm 0.1}$	46.9$_{\pm 0.2}$	63.5$_{\pm 0.2}$	13.7$_{\pm 0.1}$	38.4$_{\pm 0.2}$	65.3$_{\pm 0.1}$	35.7$_{\pm 0.2}$
⊕ALT-FLD	51.9	51.6$_{\pm 0.1}$	43.4$_{\pm 0.2}$	38.1	70.1$_{\pm 0.1}$	61.5$_{\pm 0.1}$	46.7$_{\pm 0.2}$	64.9$_{\pm 0.2}$	11.9$_{\pm 0.1}$	39.6$_{\pm 0.2}$	65.4$_{\pm 0.1}$	36.2$_{\pm 0.2}$
⊕**ALT-FLD**$_{\times 2}$	52.0	52.2$_{\pm 0.1}$	43.2$_{\pm 0.2}$	38.0	70.7$_{\pm 0.1}$	61.5$_{\pm 0.1}$	46.5$_{\pm 0.2}$	65.3$_{\pm 0.2}$	11.3$_{\pm 0.1}$	38.7$_{\pm 0.2}$	65.5$_{\pm 0.1}$	36.4$_{\pm 0.2}$

(b) LLaMA-3.1-70B.

	Avg.	Logic	Math	Code	NLI	Others	BBH (3-shot)		BBH (0-shot)		MMLU	
								CoT		CoT		Pro
LLaMA-3.1-70B	60.0	57.4$_{\pm 0.4}$	60.0$_{\pm 0.5}$	46.2	73.7$_{\pm 0.3}$	67.7$_{\pm 0.3}$	60.4$_{\pm 0.3}$	82.1$_{\pm 0.2}$	6.5$_{\pm 0.1}$	50.1$_{\pm 0.3}$	78.7$_{\pm 0.2}$	50.7$_{\pm 0.4}$
⊕ALT-PRP$_{\text{w/o RecAdam}}$	58.8	54.3$_{\pm 0.4}$	59.2$_{\pm 0.5}$	48.2	72.7$_{\pm 0.3}$	65.9$_{\pm 0.3}$	60.4$_{\pm 0.4}$	81.5$_{\pm 0.5}$	6.1$_{\pm 0.2}$	48.3$_{\pm 0.4}$	78.5$_{\pm 0.5}$	50.7$_{\pm 0.4}$
⊕ALT-PRP	60.4	57.7$_{\pm 0.4}$	59.8$_{\pm 0.5}$	49.2	73.5$_{\pm 0.3}$	67.6$_{\pm 0.3}$	60.4$_{\pm 0.4}$	82.2$_{\pm 0.3}$	6.0$_{\pm 0.2}$	50.1$_{\pm 0.4}$	78.7$_{\pm 0.3}$	50.9$_{\pm 0.4}$
⊕ALT-RT	62.7	61.4$_{\pm 0.2}$	62.1$_{\pm 0.3}$	50.8	75.4$_{\pm 0.2}$	68.4$_{\pm 0.2}$	64.1$_{\pm 0.3}$	82.5$_{\pm 0.2}$	11.5$_{\pm 0.2}$	59.2$_{\pm 0.3}$	79.0$_{\pm 0.2}$	52.4$_{\pm 0.3}$
⊕ALT-FLD	64.2	65.7$_{\pm 0.1}$	63.6$_{\pm 0.2}$	52.0	75.3$_{\pm 0.1}$	68.5$_{\pm 0.3}$	65.0$_{\pm 0.2}$	83.6$_{\pm 0.1}$	12.1$_{\pm 0.1}$	59.9$_{\pm 0.2}$	79.3$_{\pm 0.1}$	54.4$_{\pm 0.2}$
⊕**ALT-FLD**$_{\times 2}$	64.4	66.1$_{\pm 0.1}$	63.3$_{\pm 0.2}$	52.4	76.1$_{\pm 0.1}$	68.5$_{\pm 0.1}$	65.4$_{\pm 0.2}$	83.6$_{\pm 0.2}$	11.4$_{\pm 0.1}$	60.8$_{\pm 0.2}$	79.5$_{\pm 0.1}$	54.4$_{\pm 0.2}$

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Capabilities that can enhance Additional Logic Training

Logical Reasoning tasks

ALT improved LLaMA-3.1-70B by up to 30 points on logical reasoning tasks.

It also boosted **abductive reasoning**, even though only deductive logic was trained.

This shows ALT teaches **core logical structures** useful across reasoning types.

DP1 (Unknown Facts): Solved logic with unfamiliar terms → real generalization.

DP2 (Invalid Reasoning): Answered "neutral" when facts were insufficient.

DP3 (Logic Rules): Solved syllogism in a FOLIO task (e.g., eel → fish → not plant).

DP4 (Linguistic Diversity): Understood natural, Wikipedia-style expressions

(a) LLaMA-3.1-8B.

	Avg.	Logic	Math	Code	NLI	Others	BBH (3-shot)		BBH (0-shot)		MMLU	
								CoT		CoT		Pro
LLaMA-3.1-8B	47.9	$42.8_{\pm0.4}$	$39.6_{\pm0.5}$	35.4	$65.4_{\pm0.3}$	$60.7_{\pm0.3}$	$44.9_{\pm0.4}$	$61.9_{\pm0.4}$	$8.2_{\pm0.2}$	$36.5_{\pm0.4}$	$65.3_{\pm0.4}$	$35.8_{\pm0.4}$
⊕ALT-PRP	48.1	$43.7_{\pm0.2}$	$39.2_{\pm0.3}$	35.7	$65.6_{\pm0.2}$	$60.8_{\pm0.2}$	$44.9_{\pm0.2}$	$61.8_{\pm0.2}$	$8.2_{\pm0.1}$	$36.4_{\pm0.2}$	$65.3_{\pm0.2}$	$35.3_{\pm0.2}$
⊕ALT-RT	50.1	$46.8_{\pm0.1}$	$42.4_{\pm0.2}$	36.5	$68.6_{\pm0.1}$	$61.3_{\pm0.1}$	$46.9_{\pm0.2}$	$63.5_{\pm0.2}$	$13.7_{\pm0.1}$	$38.4_{\pm0.2}$	$65.3_{\pm0.1}$	$35.7_{\pm0.2}$
⊕ALT-FLD	51.9	$51.6_{\pm0.1}$	$43.4_{\pm0.2}$	38.1	$70.1_{\pm0.1}$	$61.5_{\pm0.1}$	$46.7_{\pm0.2}$	$64.9_{\pm0.2}$	$11.9_{\pm0.1}$	$39.6_{\pm0.2}$	$65.4_{\pm0.1}$	$36.2_{\pm0.2}$
⊕**ALT-FLD**$_{\times2}$	52.0	$52.2_{\pm0.1}$	$43.2_{\pm0.2}$	38.0	$70.7_{\pm0.1}$	$61.5_{\pm0.1}$	$46.5_{\pm0.2}$	$65.3_{\pm0.2}$	$11.3_{\pm0.1}$	$38.7_{\pm0.2}$	$65.5_{\pm0.1}$	$36.4_{\pm0.2}$

(b) LLaMA-3.1-70B.

	Avg.	Logic	Math	Code	NLI	Others	BBH (3-shot)		BBH (0-shot)		MMLU	
								CoT		CoT		Pro
LLaMA-3.1-70B	60.0	$57.4_{\pm0.4}$	$60.0_{\pm0.5}$	46.2	$73.7_{\pm0.3}$	$67.7_{\pm0.3}$	$60.4_{\pm0.3}$	$82.1_{\pm0.2}$	$6.5_{\pm0.1}$	$50.1_{\pm0.3}$	$78.7_{\pm0.3}$	$50.7_{\pm0.4}$
⊕ALT-PRP	60.4	$57.7_{\pm0.4}$	$59.8_{\pm0.5}$	49.2	$73.5_{\pm0.3}$	$67.6_{\pm0.3}$	$60.4_{\pm0.4}$	$82.2_{\pm0.3}$	$6.0_{\pm0.2}$	$50.1_{\pm0.4}$	$78.7_{\pm0.3}$	$50.9_{\pm0.4}$
⊕ALT-RT	62.7	$61.4_{\pm0.2}$	$62.1_{\pm0.3}$	50.8	$75.4_{\pm0.2}$	$68.4_{\pm0.2}$	$64.1_{\pm0.3}$	$82.5_{\pm0.2}$	$11.5_{\pm0.2}$	$59.2_{\pm0.3}$	$79.0_{\pm0.2}$	$52.4_{\pm0.3}$
⊕ALT-FLD	64.2	$65.7_{\pm0.1}$	$63.6_{\pm0.2}$	52.0	$75.3_{\pm0.1}$	$68.5_{\pm0.1}$	$65.0_{\pm0.2}$	$83.6_{\pm0.1}$	$12.1_{\pm0.1}$	$59.9_{\pm0.2}$	$79.3_{\pm0.1}$	$54.4_{\pm0.2}$
⊕**ALT-FLD**$_{\times2}$	64.4	$66.1_{\pm0.1}$	$63.3_{\pm0.2}$	52.4	$76.1_{\pm0.1}$	$68.5_{\pm0.1}$	$65.4_{\pm0.2}$	$83.6_{\pm0.2}$	$11.4_{\pm0.1}$	$60.8_{\pm0.2}$	$79.5_{\pm0.1}$	$54.4_{\pm0.2}$

Table 3: LLaMA-3.1-8B trained on the ablation corpora.

	Avg.	Logic	Math	Code	NLI	Others	BBH (3-shot)		BBH (0-shot)		MMLU	
								CoT		CoT		Pro
⊕**ALT-FLD**$_{\times2}$	**52.0**	$\mathbf{52.2}_{\pm0.1}$	$\mathbf{43.2}_{\pm0.2}$	38.0	$70.7_{\pm0.1}$	$\mathbf{61.5}_{\pm0.1}$	$46.5_{\pm0.2}$	$\mathbf{65.3}_{\pm0.2}$	$11.3_{\pm0.1}$	$38.7_{\pm0.2}$	$65.5_{\pm0.1}$	$\mathbf{36.4}_{\pm0.2}$
w/o DP1	51.4	$\mathbf{52.2}_{\pm0.1}$	$43.1_{\pm0.2}$	**39.2**	$70.0_{\pm0.1}$	$59.4_{\pm0.1}$	$46.7_{\pm0.2}$	$64.7_{\pm0.2}$	$11.5_{\pm0.1}$	$38.9_{\pm0.2}$	$65.4_{\pm0.1}$	$36.1_{\pm0.2}$
w/o DP2	50.6	$49.9_{\pm0.1}$	$43.1_{\pm0.2}$	38.1	$\mathbf{71.1}_{\pm0.1}$	$59.3_{\pm0.1}$	$46.1_{\pm0.2}$	$64.6_{\pm0.2}$	$10.4_{\pm0.1}$	$37.4_{\pm0.2}$	$65.4_{\pm0.1}$	$35.7_{\pm0.2}$
w/o DP3.rules	50.7	$50.4_{\pm0.1}$	$42.8_{\pm0.2}$	38.3	$69.5_{\pm0.1}$	$59.4_{\pm0.1}$	$46.4_{\pm0.2}$	$64.0_{\pm0.2}$	$11.8_{\pm0.1}$	$38.3_{\pm0.2}$	$\mathbf{65.6}_{\pm0.1}$	$36.2_{\pm0.2}$
w/o DP3.steps	51.1	$51.5_{\pm0.1}$	$43.1_{\pm0.2}$	38.7	$69.6_{\pm0.1}$	$59.5_{\pm0.1}$	$\mathbf{46.8}_{\pm0.2}$	$65.0_{\pm0.2}$	$12.3_{\pm0.1}$	$38.8_{\pm0.2}$	$\mathbf{65.6}_{\pm0.1}$	$36.3_{\pm0.2}$
w/o DP4	51.3	$\mathbf{52.2}_{\pm0.1}$	$42.8_{\pm0.2}$	38.4	$70.3_{\pm0.1}$	$59.5_{\pm0.1}$	$46.1_{\pm0.2}$	$64.8_{\pm0.2}$	$\mathbf{12.8}_{\pm0.1}$	$\mathbf{39.3}_{\pm0.2}$	$65.5_{\pm0.1}$	$36.3_{\pm0.2}$

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Capabilities that can enhance Additional Logic Training

Math tasks

ALT boosted LLaMA-3.1-70B by:

+7 points on math tasks.

+10 points on coding tasks.

Reasoning for Math Gains:

Mastery of **predicate logic** (taught by ALT) helps solve math problems.

Coding tasks

Previous studies showed the **reverse is also true** — training on **code improves logical reasoning**.

(b) Math.

	GSM8k			MATH	MathQA
		CoT	CoT (0-shot)	-	-
LLaMA-3.1-70B	$80.9_{\pm 1.1}$	$75.2_{\pm 1.2}$	$65.4_{\pm 1.3}$	$23.7_{\pm 0.6}$	$55.0_{\pm 0.9}$
⊕**ALT-FLD**$_{\times 2}$	$\mathbf{83.3}_{\pm 0.4}$	$\mathbf{80.4}_{\pm 0.4}$	$\mathbf{73.0}_{\pm 0.5}$	$\mathbf{24.4}_{\pm 0.2}$	$\mathbf{55.4}_{\pm 0.4}$

(c) Code.

	HumanEval	MBPP	MBPP+	MultiPL-E (cpp)	MultiPL-E (go)
LLaMA-3.1-70B	32.3	43.4	48.7	29.8	76.6
⊕**ALT-FLD**$_{\times 2}$	**42.6**	**49.5**	**52.5**	**38.7**	**78.6**

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Benchmark Performance Breakdown

Logic Tasks

- LogicNLI: 34.9% → 50.9%
- FOLIO: 58.9% → 66.7%
- RobustLR: 49.6% → 81.6%

Math and Code

- GSM8k (CoT): 80.9% → 83.3%
- MathQA: 65.4% → 73.0%
- HumanEval: 32.3% → 42.6%

NLI & Others

- MNLI: 82.2% → 83.7%
- RTE: 84.0% → 87.2%
- CommonsenseQA: 81.2% → 82.5%

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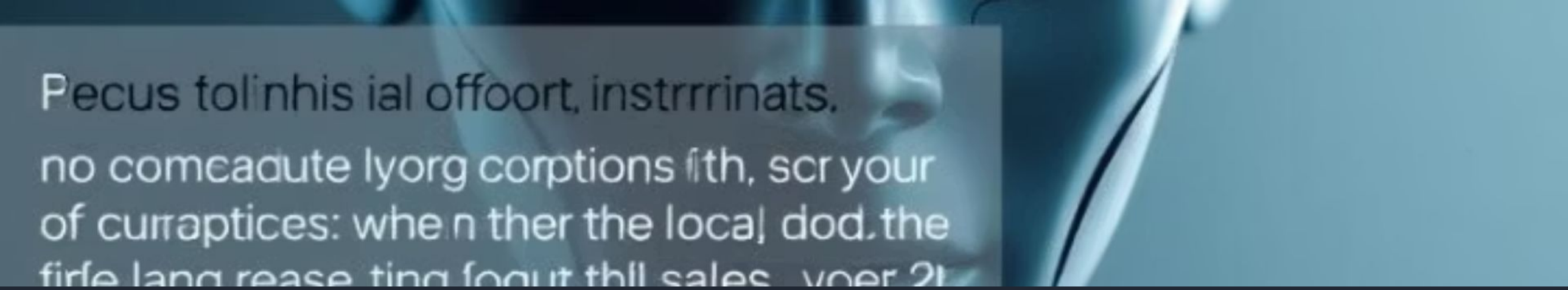

Case Analysis and Examples

DP1: Handling Unknown Facts

Models reason correctly with unknown or nonsensical entities.

DP2: Detecting Illogical Reasoning

Models reject invalid or insufficient premises.

DP3: Application of Diverse Rules

Employ syllogistic logic to derive valid conclusions.

DP4: Linguistic Expression Robustness

Process naturalistic prompts akin to Wikipedia style.

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Future Work

- Combine ALT with domain-specific factual knowledge for broader coverage.
- Fine-tune logic-enhanced LLMs on specialized tasks (e.g., law, science).
- Explore performance scaling on larger models (>70B parameters).
- Use ALT-trained models to improve few-shot and zero-shot capabilities.
- Develop interpretable AI systems that can explain logical steps.
- Integrate ALT into curriculum learning pipelines to gradually build reasoning skills.

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Conclusion

- Our synthetic logic corpus significantly enhances LLaMA models' reasoning abilities.
- Design principles ensure robustness in handling unknown facts and illogical premises.
- Empirical results demonstrate cross-domain improvements in logic, math, code, and NLI tasks.
- Future work aims to integrate logic-enhanced models into specialized domains and scaling efforts.

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